

Qatar-2b

R-2009

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_140627 · created 2026-08-07T16:56:44+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456835.694879 +/- 5.7e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.001 +/- 0.066
Transit depth (Rp/Rs)^2	0.00012 +/- 0.015 [%]
Orbital Inclination (inc)	88.73 +/- 1.77
Airmass coefficient 1 (a1)	1.41 +/- 1.08
Airmass coefficient 2 (a2)	-0.29 +/- 0.92
Scatter in the residuals of the lightcurve fit is	81.53 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.75
Transit Duration (day)	0.0562 +/- 0.0021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.001 ± 0.066 vs 0.16327 (deviation 2.46σ)
Duration fit vs published	0.0562 d vs 0.0767 d (deviation 9.75σ)
Mid-time O–C	0.2 min
Depth SNR	0.0
Residual scatter	81.53 %

Light curve

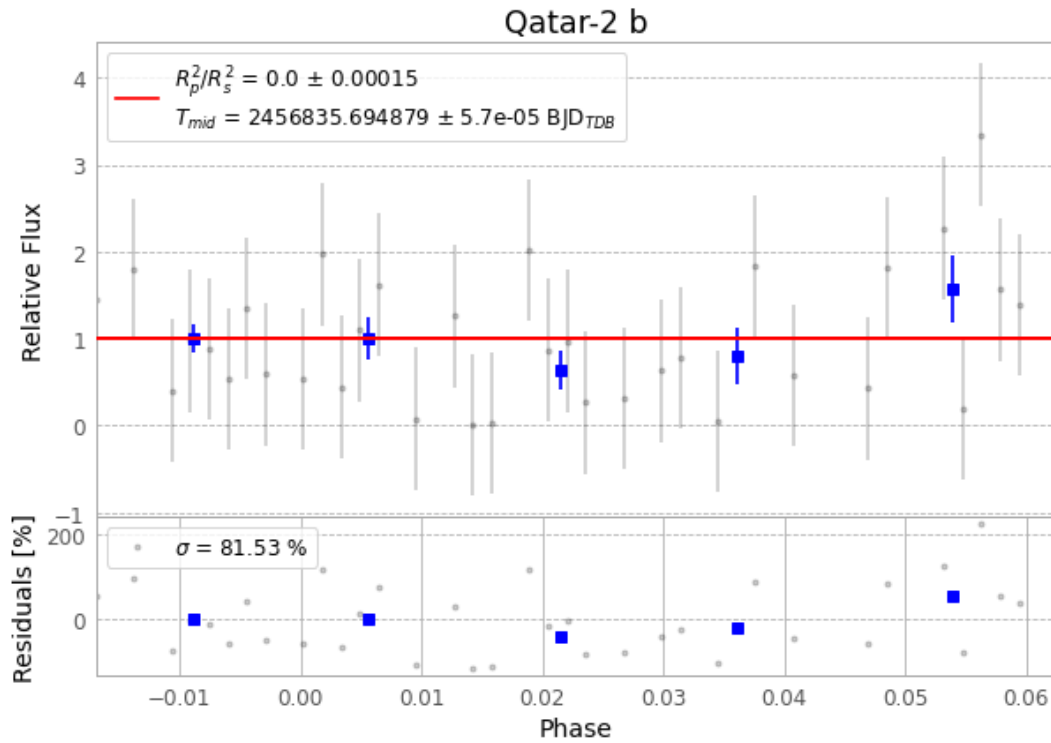


Figure 1 — FinalLightCurve_Qatar-2 b_2014-06-26.png (SHA-256 f808ac8a2029a004...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [431, 472], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c25a0c24641bc08a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

52 FITS frames (34 MB), Qatar - 2140627040313.FITS → Qatar - 2140627063612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-140627.log (35c29e9a461e...), FinalLightCurve_Qatar-2 b_2014-06-26.png (f808ac8a2029...), FinalLightCurve_Qatar-2 b_2014-06-26.csv (264e79e94388...)

Compute resources

Wall time 107.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:56Z.